

Abstract Algebra – Dummit & Foote

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Contents

I	Group Theory	2
1	Introduction to Groups	2
1.1	Axioms	2
1.2	Dihedral Groups	3
1.3	Symmetric Groups	3
1.4	Matrix Groups	3
1.5	The Quaternion Group	4
1.6	Homomorphisms and Isomorphisms	4
1.7	Group Actions	5
2	Subgroups	5
2.1	Centralizers and Normalizers, Stabilizers and Kernels	5
2.2	Cyclic Subgroups	6
2.2.1	Theorems	7
2.3	Subgroups Generated by Subsets	7
3	Quotient Groups and Homomorphisms	8
3.1	Definitions	8
3.2	Theorems	9
3.2.1	Quotient Groups from Kernel	9
3.2.2	Quotient Groups from Normal Subgroups	10
3.3	Lagrange's Theorem	11
3.3.1	Statement	11
3.3.2	Corollaries	11
3.3.3	Cauchy's Theorem and Sylow's Theorem	12
3.3.4	Applications: Fermat's Little Theorem, Euler's Theorem	12
3.4	Isomorphism Theorems	12
3.5	The Hölder Program	13
3.5.1	Composition Series	13
3.5.2	Hölder Program	13
3.6	The Alternating Group A_n	14
4	Group Actions	14

4.1	Definitions	14
4.2	Groups Acting on Themselves by Conjugation	15
4.2.1	Definitions	15
4.2.2	Conjugacy in Symmetric Groups	16
4.3	Automorphisms	17
4.4	Sylow Theorems	17
4.4.1	Statement	17
4.4.2	Applications	18
5	Direct and Semidirect Products and Abelian Groups	19
5.1	Direct Products	19
5.2	Fundamental Theorem of Finitely-Generated Abelian Groups	19

Part I

Group Theory

1 Introduction to Groups

1.1 Axioms

If $\star$ is a function on set G such that $\star : G \times G \rightarrow G$, then $\star$ is a **binary operation** on G .

- Associative if $\forall a, b \in G, a \star (b \star c) = (a \star b) \star c$
- Commutative if $\forall a, b \in G, a \star b = b \star a$

A **group** is an ordered pair $(G, \star)$ such that

1. $\star$ is associative
2. Existence of identity: $\exists e \in G$ such that $\forall a \in G, a \star e = e \star a = a$
3. Existence of inverse: $\forall a \in G, \exists a^{-1} \in G$ such that $a \star a^{-1} = a^{-1} \star a = e$

*A group is **Abelian (commutative)** if $\forall a, b \in G, a \star b = b \star a$

E.g. $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ are groups under $+$ with identity $e = 0$ and inverse $a^{-1} = -a$. But $\mathbb{Z} - \{0\}$ is not a group under $\times$ because inverses are not guaranteed to exist.

Let $(G, \star)$ be a group, and $x \in G$. The **order** of x is the smallest positive integer n such that $x^n = 1$, and is denoted $|x|$. If no positive powers of x is 1, $|x| = \infty$.

- In the additive group $\mathbb{Z}/9\mathbb{Z}$, $\bar{6}$ has order 3, since $\bar{6} + \bar{6} + \bar{6} = \bar{18} = \bar{0}$.
- In the multiplicative group $\mathbb{R} - \{0\}$, -1 has order 2 since $(-1)^2 = 1$.

1.2 Dihedral Groups

For each $n \in \mathbb{Z}^+$, $n \geq 3$, let D_{2n} be the set of symmetries of a regular n -gon. E.g. $s \in D_{2n}$ might be a rotation of $2\pi/n$ around the n -gon.

$|D_{2n}| = 2n$, so we call D_{2n} the **dihedral group of order $2n$** .

Any symmetry can be written in terms of rotations (r) and reflections (s) alone, e.g. for $n = 12$,

$$(sr^9)(sr^6) = s(r^9s)r^6 = s(sr^{-9})r^6 = s^2r^{-3} = r^9$$

S is a set of generators of G if any element of G can be written as a finite product of the elements of S . This is notated:

$$G = \langle S \rangle$$

For the dihedral group, $S = \{s, r\}$. The equations in G that the generators satisfy are called **relations** in G . For D_{2n} , the relations are $r^n = 1$, $s^2 = 1$, and $rs = sr^{-1}$.

If G is generated by S , and there exists a collection of relations $R_1, R_2, \dots, R_m$, then these generators and relations are called the **presentation of G** :

$$G = \langle S \mid R_1, R_2, \dots, R_m \rangle$$

For the dihedral group, $D_{2n} = \langle r, s \mid r^n = s^2 = 1, rs = sr^{-1} \rangle$.

1.3 Symmetric Groups

Let S_Ω be the bijection from Ω to itself (permutation). S_Ω is a group under the function composition $\circ$.

- Call $(S_\Omega, \circ)$ the **symmetric group** on set Ω .
- If $\Omega = \{1, 2, 3, \dots, n\}$, then we denote S_Ω as S_n , the **symmetric group of degree n** .
- $(S_n, \circ)$ has degree $n!$ ($n!$ elements).

We can represent S_n in a cycle decomposition, grouping disjoint cycles. E.g.

$$(a_1 \ a_2 \ \dots \ a_m)(a_{m_1+1} \ a_{m_1+2} \ \dots \ a_{m_2}) \cdots (a_{m_{k-1}+1} \ a_{m_{k-1}+2} \ \dots \ a_{m_k})$$

Meaning that a_1 goes to a_2 , a_{m_1+1} goes to a_{m_1+2} , etc. Values such that $\sigma(a_i) = a_i$ aren't written. For example, for $\sigma \in S_3$ with $\sigma(1) = 1$, $\sigma(2) = 3$, and $\sigma(3) = 2$, the cycle decomposition is

$$(2 \ 3)$$

1.4 Matrix Groups

First, we need to define a field. A field is a set F with 2 binary operators $+$ and $\cdot$ such that

1. $(F, +)$ and $(F - \{0\}, \cdot)$ are Abelian groups
2. The distribution law holds:

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c) \quad \forall a, b, c \in F$$

Define the **general linear group of degree n** :

$$GL_n(F) = \{A \mid A \text{ is a } n \times n \text{ matrix with entries from } F, \text{ and } \det(A) \neq 0\}$$

- Since $\det(A) = 0$, $GL_n(F)$ is closed under matrix multiplication.
- Matrix products are associative.
- $\det(A) \neq 0$ iff. A has an inverse.

Thus, $GL_n(F)$ is a group under matrix multiplication.

1.5 The Quaternion Group

Simple enough:

$$\begin{aligned} Q_8 &= \{1, -1, i, -i, j, -j, k, -k\} \\ 1 \cdot a &= a \cdot 1 = a \\ (-1) \cdot a &= a \cdot (-1) = -a \\ i \cdot i &= j \cdot j = k \cdot k = -1 \\ i \cdot j &= k, \quad j \cdot k = i, \quad k \cdot i = j \\ j \cdot i &= -k, \quad k \cdot j = -i, \quad i \cdot k = -j \end{aligned}$$

$(Q_8, \cdot)$ is a non-Abelian group of order 8.

1.6 Homomorphisms and Isomorphisms

Let $(G, \star)$ and $(H, \diamond)$ be groups. A map $\phi : G \rightarrow H$ such that

$$\phi(x \star y) = \phi(x) \diamond \phi(y)$$

is a **homomorphism**.

- “A map that reflects the group structure of the domain and codomain.”

$\phi : G \rightarrow H$ is called an **isomorphism** if

1. ϕ is a homomorphism
2. ϕ is a bijection

Can also say: G and H are isomorphic; G and H are of the same isomorphism type; $G \cong H$.

I.e. G and H are the “same group” except the elements and operators are written differently (think abstract).

If $\phi : G \rightarrow H$ is an isomorphism, then

1. $|G| = |H|$
2. G is Abelian iff. H is Abelian
3. $\forall x \in G, |x| = |\phi(x)|$

1.7 Group Actions

A group action from group G on set A is a map from $G \times A$ to A satisfying

1. $g_1 \cdot (g_2 \cdot a) = (g_1 g_2) \cdot a \quad \forall g_1, g_2 \in G, \forall a \in A$
2. $1 \cdot a = a \quad \forall a \in A$

Let G act on A . For each $g \in G$, define σ_g to be the map

$$\begin{aligned}\sigma_g : A &\rightarrow A \\ \sigma_g(a) &= g \cdot a\end{aligned}$$

1. For each $g \in G$, σ_g is a permutation of A .
2. The map from G to S_A defined by $g \mapsto \sigma_g$ is a homomorphism.

I.e. every $g \in G$ acts as a permutation of A in a way consistent with the group operation in G .

Group actions treat $g \in G$ to be functions, sort of.

2 Subgroups

A subset H of a group G is a **subgroup** of G if

1. $H \neq \emptyset$
2. H is closed under products and inverses. I.e. $x, y \in H$ implies $x^{-1} \in H$ and $xy \in H$.

This is denoted $H \leq G$, or if $H \neq G$, $H < G$.

I.e. a subgroup is a subset of G that is also a group w.r.t the binary operation defined on G . A subgroup must thus contain the identity operator.

Every G has two subgroups: $H = G$ and $H = \{1\}$ (trivial subgroup).

A simpler subgroup criterion: a subset H of a group G is a subgroup of G iff.

1. $H \neq \emptyset$
2. $\forall x, y \in H, xy^{-1} \in H$

2.1 Centralizers and Normalizers, Stabilizers and Kernels

Let A be a subset of G .

The **centralizer of A in G** is the set of elements of G that commute with every element of A .

$$C_G(A) = \{g \in G \mid gag^{-1} = a, \forall a \in A\}$$

- For an Abelian group G , $C_G(A) = G, \forall A \subseteq G$.
- For the quaternion group, $C_{Q_8}(\{i\}) = \{\pm 1, \pm i\}$

The **center** of G is the set of elements of G commuting with every element of G .

$$Z(G) = C_G(G) = \{g \in G \mid gx = xg, \forall x \in G\}$$

The normalizer of A in G is the set of elements of G that commutes with the entire set A .

$$N_G(A) = \{g \in G \mid gAg^{-1} = A\}$$

This is a looser criterion than the centralizer, since order within A doesn't matter. Formally, if $g \in C_G(A)$, then $gag^{-1} = a$ for all $a \in A$, so $C_G(A) \leq N_G(A)$.

For an Abelian group G , which has the commutation property already, $C_G(A) = N_G(A) = Z(G) = G$, for any subset $A \subseteq G$.

If G is a group action on set S , and $s \in S$, the **stabilizer of s in G** is

$$G_s = \{g \in G \mid g \cdot s = s\}$$

- $1 \in G_s$
- If $y \in G_s$, then $y^{-1} \in G_s$.

$$\begin{aligned} s &= 1 \cdot s = (y^{-1}y) \cdot s \\ &= y^{-1} \cdot (y \cdot s) \\ &= y^{-1} \cdot s \end{aligned}$$

- If $x, y \in G_s$, then $xy \in G_s$.

$$\begin{aligned} (xy) \cdot s &= x \cdot (y \cdot s) \\ &= x \cdot s \\ &= s \end{aligned}$$

Thus G_s is a subgroup of the group action G , or $G_s \leq G$.

The **kernel of the action of G on set S** is

$$\{g \in G \mid g \cdot s = s, \forall s \in S\}$$

2.2 Cyclic Subgroups

A possible way to generate a subgroup H of G from an element $x \in G$ is to form the set $H = \{x^n \mid n \in \mathbb{Z}\}$. Guarantees closure under products and inverses.

A subgroup $H \subseteq G$ is **cyclic** if $\exists x \in H$ such that $H = \{x^n \mid n \in \mathbb{Z}\}$. I.e. H can be generated by 1 element. Notated $H = \langle x \rangle$. Cyclic groups are Abelian, by properties of the exponent.

- E.g. If $H = (\mathbb{Z}, +)$, then $H = \langle 1 \rangle$.
- E.g. If H is the subgroup of all rotations of an n -gon, then $H = \langle r \rangle$.

2.2.1 Theorems

1. If $H = \langle x \rangle$, then $|H| = |x|$.
2. Let $x \in G$. For $m, n \in \mathbb{Z}$, if $x^n = x^m = 1$, then $x^d = 1$, where $d = \gcd(m, n)$.
3. Any 2 cyclic groups of the same order are isomorphic.
 - If $n \in \mathbb{Z}^+$ and $\langle x \rangle, \langle y \rangle$ are cyclic groups of order n , the following map is an isomorphism.

$$\begin{aligned}\phi : \langle x \rangle &\rightarrow \langle y \rangle \\ x^k &\mapsto y^k\end{aligned}$$

- If $\langle x \rangle$ is an infinite cyclic group, the following map is a homomorphism. Thus for each $n \in \mathbb{Z}^+$, let Z_n be the cyclic group of order n .

$$\begin{aligned}\phi : \mathbb{Z} &\rightarrow \langle x \rangle \\ k &\mapsto x^k\end{aligned}$$

4. Let G be a map, $x \in G$, and $a \in \mathbb{Z} - \{0\}$.
 - If $|x| = \infty$, then $|x^a| = \infty$
 - If $|x| = n < \infty$, then $|x^a| = n / \gcd(n, a)$
5. Let $H = \langle x \rangle$.
 - If $|x| = \infty$, then $H = \langle x \rangle$ iff. $a = \pm 1$.
 - If $|x| = n < \infty$, then $H = \langle x^a \rangle$ iff. $\gcd(a, n) = 1$. The number of generators of H is $\phi(n)$.
 - E.g. $\bar{a}$ generates $\mathbb{Z}/n\mathbb{Z}$ iff. $(\bar{a}, n) = 1$. So $\bar{1}, \bar{5}, \bar{7}, \bar{11}$ are generators of $\mathbb{Z}/12\mathbb{Z}$, and these are all the $\phi(12) = 4$ generators.
6. Let $H = \langle x \rangle$ be a cyclic group. Then
 - Every subgroup of H is cyclic.
 - If $|H| = \infty$, then for any non-negative and distinct $a, b \in \mathbb{Z}$, $\langle x^a \rangle = \langle x^b \rangle$.
 - If $|H| = n < \infty$, then for each $a \in \mathbb{Z}^+$ such that $a|n$, there's a unique subgroup of order a . This is the cyclic group $\langle x^d \rangle$, where $d = n/a$.

2.3 Subgroups Generated by Subsets

If $\mathcal{A}$ is any nonempty collection of subgroups of G , the intersection of all $\mathcal{A}$ is also a subgroup of G .

$$K = \bigcap_{H \in \mathcal{A}} H \text{ is a group if } H \leq G$$

If A is a subset of the group G , let the **subgroup of G generated by A** be

$$\langle A \rangle = \bigcap_{A \subseteq H, H \leq G} H$$

I.e. the intersection of all subgroups of G containing A . Equivalently,

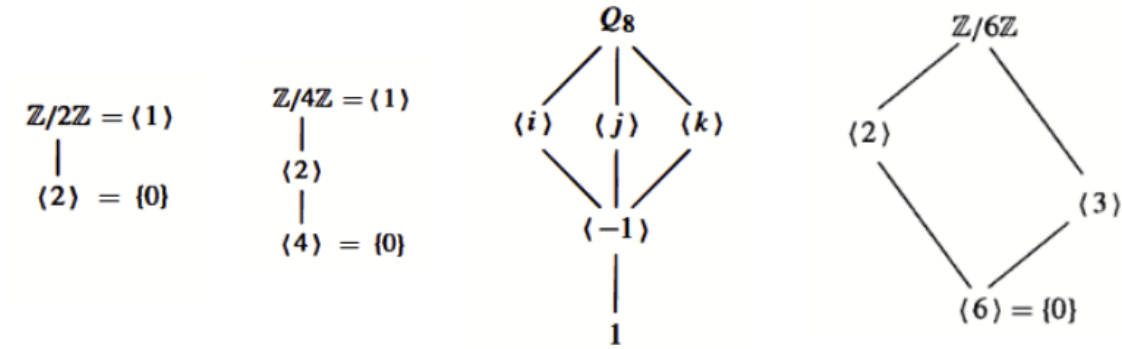
$$\langle A \rangle = \{a_1^{\alpha_1} a_2^{\alpha_2} \cdots a_n^{\alpha_n} \mid \forall i, a_i \in A, \alpha_i \in \mathbb{Z}, a_i \neq a_{i+1} \text{ and } n \in \mathbb{Z}^+\}$$

An Abelian group G simplifies this a bit:

$$\langle A \rangle = \{a_1^{\alpha_1} a_2^{\alpha_2} \cdots a_n^{\alpha_n} \mid \forall i \text{ and } \alpha_i \in \mathbb{Z}\}$$

- Without an Abelian structure, determining the generated subgroup becomes difficult. E.g. if $G = D_8$, let r, s be the generators. Let $a = s, b = rs$ so that $G = \langle a, b \rangle$. Thus G is generated by 2 elements of order 2, but has order 8.
- Or for $G = S_n = \langle (1\ 2), (1\ 2\ 3 \dots n) \rangle$. Generated by elements of order 2 and n , but has order $n!$

Lattice diagrams can help illuminate the structure of groups and subgroups. Build up from $\{0\}$ at the bottom, and draw subgroups of increasingly higher order upwards. This is helpful to see, for example, the smallest subgroup of G , $\langle H, K \rangle$, containing two smaller subgroups $H \leq G$ and $K \leq G$.

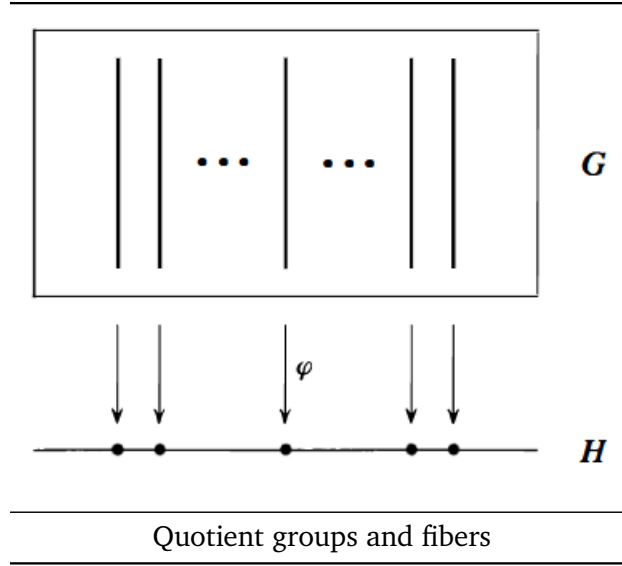


Subgroup examples

3 Quotient Groups and Homomorphisms

3.1 Definitions

Quotient groups are a way to obtain a smaller group from a group G .



$\phi : G \rightarrow H$ is a homomorphism from G to H , i.e. $\phi(xy) = \phi(x)\phi(y)$ for $x, y \in G$. Vertical lines above each point $h \in H$ represent the **fiber of ϕ over h** .

Denote the **kernel of ϕ** as the set

$$\ker \phi = \{g \in G \mid \phi(g) = 1_H\}$$

Let K be the kernel. The **quotient group/factor group** G/K (" $G \bmod K$ ") is a group whose elements are the fibers of ϕ .

For any subgroup $N \leq G$ and $g \in G$, define

$$\begin{aligned} \text{Left coset of } N \text{ in } G: \quad gN &= \{gn \mid n \in N\} \\ \text{Right coset of } N \text{ in } G: \quad Ng &= \{ng \mid n \in N\} \end{aligned}$$

Call any element of a coset the **representative** of that coset. The fibers of a homomorphism $\phi : G \rightarrow H$ are the left or right cosets of the kernel (it may help to think of this graphically). The elements of G/K are the left cosets gK or the right cosets Kg with $g \in G$.

3.2 Theorems

3.2.1 Quotient Groups from Kernel

Theorem 1: Let $\phi : G \rightarrow H$ be a homomorphism. Then

1. $\phi(1_G) = 1_H$
2. $\phi(g^{-1}) = \phi(g)^{-1}$
3. $\phi(g^n) = \phi(g)^n$
4. Kernel of ϕ is a subgroup of G : $\ker \phi \leq G$

5. Image of G under ϕ is a subgroup of H : $\text{im}(\phi) \leq H$

Proof: For (1), $\phi(1_G) = \phi(1_G 1_G) = 1_H 1_H$. For (2), $\phi(g)\phi(g^{-1}) = \phi(gg^{-1}) = \phi(1_G) = 1_H$. Then (3) follows from induction.

For (4), note that $1_G \in \ker \phi$, so $\ker \phi$ is nonempty. Let $x, y \in \ker \phi$. Then $\ker \phi$ satisfies the subgroup criterion:

$$\phi(xy^{-1}) = \phi(x)\phi(y)^{-1} = 1_H$$

Similarly for (5), note that $\phi(1_G) = 1_H$, so the identity of H lies in $\text{im}(\phi)$. If $x, y \in \text{im}(\phi)$ with $\phi(a) = x$ and $\phi(b) = y$, then $\text{im}(\phi)$ satisfies the subgroup criterion:

$$\begin{aligned} xy^{-1} &= \phi(a)\phi(b)^{-1} = \phi(ab^{-1}) \\ xy^{-1} &\in \text{im}(\phi) \end{aligned}$$

Definition: Let $\phi : G \rightarrow H$ with kernel K . Then the **quotient group** G/K is the group whose elements are fibers of ϕ with group operations defined such that: if X is the fiber above a and Y is the fiber above b , then $X \cdot Y$ is the fiber above ab .

Thus $K \in G/K$. Other elements of G/K are “translations” of K .

Theorem 2: Let $\phi : G \rightarrow H$ be a homomorphism with kernel K . Let $X \in G/K$ be a fiber above a , i.e. $X = \phi^{-1}(a)$. Then

- 1) $\forall u \in X, X = \{uk \mid k \in K\}$
- 2) $\forall u \in X, X = \{ku \mid k \in K\}$

Proof: First show $uK \subseteq X$, then $X \subseteq uK$. Similarly for Ku .

What this shows is that the fibers of a homomorphism are the left (or right) cosets of the kernel. Furthermore, the specific value of u doesn't matter, as long as $u \in X$. This is why u is called the “representative.”

Theorem 3: Let K be the kernel of $\phi : G \rightarrow H$. The set of left cosets of G with operation defined as

$$uK \cdot vK = (uv)K$$

form the quotient group G/K . Similarly for right cosets.

E.g. for $\phi : \mathbb{Z} \rightarrow \mathbb{Z}_n$, the cosets are $a + n\mathbb{Z}$ and the kernel is $n\mathbb{Z}$. Thus the fibers $\mathbb{Z}/n\mathbb{Z}$ form a group.

In essence, to find a quotient group:

1. Find homomorphism ϕ from group G .
2. Find kernel K of homomorphism.
3. G/K , the fibers of ϕ , is a group

3.2.2 Quotient Groups from Normal Subgroups

We can form a group from a subgroup N iff. N is the kernel of some homomorphism. To determine whether N is a kernel of such a homomorphism, check whether N is a normal subgroup: $N \trianglelefteq G$.

Theorem: For a subgroup N of G , i.e. $N \leq G$, the following are equivalent conditions.

1. $N \trianglelefteq G$
2. $N_G(N) = G$
3. $gN = Ng, \quad \forall g \in G$
4. $gNg^{-1} \subseteq N, \quad \forall g \in G$

Theorem: $N \leq G$ is normal iff. it's the kernel of some homomorphism.

Proof: If N is the kernel of some homomorphism, its left and right cosets are equal (both are fibers of ϕ). By the previous theorem, if $gN = Ng$ for all $g \in G$ is equivalent to $N \trianglelefteq G$.

In the opposite direction, assume $N \trianglelefteq G$ and let $\pi : G \rightarrow G/N$ with $\pi(g) = gN$. Then

$$\pi(g_1g_2) = g_1g_2N = (g_1N)(g_2N) = \pi(g_1)\pi(g_2)$$

so π is a homomorphism. The kernel is $\ker \pi = \{g \in G \mid gN = 1N\} = N$, the normal subgroup.

For $N \trianglelefteq G$, the homomorphism $\pi : G \rightarrow G/N$ defined by $\pi(g) = gN$ is the **natural projection homomorphism**.

Thus an easy way to see whether a subgroup $N \leq G$ is a kernel of some homomorphism is to check whether $N_G(N) = G$. Then we can find the quotient group G/N .

Note that in an Abelian group, any subgroup $N \leq G$ is normal, so any $N \leq G$ can be used to create a quotient group G/N .

3.3 Lagrange's Theorem

3.3.1 Statement

Theorem: The left cosets of $N \leq G$ form a partition of G , and $uN = vN$ iff. u, v are representatives of the same coset.

Lagrange's Theorem: If G is a finite group and $H \leq G$, then $|H|$ divides $|G|$. The number of left cosets of H in G is $|G|/|H|$.

Proof: Let $|H| = n$ and k be the number of left cosets of H in G . For some $g \in G$, define $\phi : H \rightarrow gH$ with $\phi(h) = gh$. ϕ is both injective and surjective, so $|gH| = |H| = n$. From the previous theorem, the cosets partition G , so $|G| = k|H| = kn$. Thus $|G|/|H| = k$.

If $H \leq G$, the number of left cosets of H in G is called the **index** of H in G , denoted $|G : H|$.

- E.g. $\{0\} \in \mathbb{Z}$ has infinite index.
- E.g. $\langle n \rangle \in \mathbb{Z}$ has index n .

3.3.2 Corollaries

Corollary 1: If $x \in G$ and $|G|$ is finite, then $|x|$ divides $|G|$, i.e. $x^{|G|} = 1$.

Proof: $|x| = |\langle x \rangle|$. Let $H = \langle x \rangle$. Then $H \leq G$ so $|H| = |x|$ divides $|G|$.

Corollary 2: If $|G|$ is prime, then G is cyclic and $G \cong \mathbb{Z}_p$.

Proof: Let $x \in G$ and $x \neq 1$. Then $|\langle x \rangle| \geq 1$. $\langle x \rangle \in G$, so $|\langle x \rangle|$ divides $|G|$. But $|G|$ is prime, so $|G| = |\langle x \rangle|$. Hence G is cyclic with order p .

Corollary 3: Let $H \leq G$ and $|G : H| = 2$. Then $H \trianglelefteq G$.

Proof: Let $g \in G - H$. The two left cosets of H in G are $1H$ and gH . They partition G , so $gH = G - H$. Similarly, $Hg = g - H$. Then $gH = Hg$, so H must be a normal subgroup of G .

- E.g. $\langle i \rangle = \{i, -1, -i, +1\}$, which has order 4. Q_8 has order 8, so $|Q_8 : \langle i \rangle| = 2$. Thus $\langle i \rangle \trianglelefteq Q_8$.

3.3.3 Cauchy's Theorem and Sylow's Theorem

The converse to Lagrange's theorem isn't true generally: just because n divides G doesn't mean there exists $H \leq G$ with $|H| = n$.

There are some partial converses:

Cauchy's Theorem: If G is finite, and there exists a prime p dividing $|G|$, then G has an element of order p .

Sylow's Theorem: If G is a finite group of order $p^\alpha m$, where p is prime and doesn't divide m , then G has a subgroup of order p^α .

3.3.4 Applications: Fermat's Little Theorem, Euler's Theorem

Lagrange's theorem provides an elegant way to prove these (seemingly number-theoretic) theorems.

Fermat's Little Theorem: If p is prime, then $a^p \equiv a \pmod{p}$ for all $a \in \mathbb{Z}$.

Proof: Define the multiplicative group modulo p , i.e. $(\mathbb{Z}/p\mathbb{Z})^\times$. For all $a \in \mathbb{Z}$ (with operation being multiplication mod p), $\langle a \rangle \in (\mathbb{Z}/p\mathbb{Z})^\times$. Thus $|a| = |\langle a \rangle|$ must divide $p - 1$, so $a^{p-1} \equiv 1 \pmod{p}$.

Euler's Theorem: For all a relatively prime to n , $a^{\phi(n)} \equiv 1 \pmod{n}$, where ϕ is the Euler totient function.

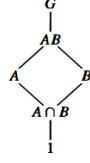
Proof: Define the multiplicative group mod n , $(\mathbb{Z}/n\mathbb{Z})^\times$. Also define the multiplicative group mod n of elements *relatively prime* to n , $(\mathbb{Z}/n\mathbb{Z})^*$. Then $|(\mathbb{Z}/n\mathbb{Z})^*| = \phi(n)$. Let a be relatively prime to n , so $a \in (\mathbb{Z}/n\mathbb{Z})^*$. Then all powers of a are also relatively prime to n , so $\langle a \rangle \in (\mathbb{Z}/n\mathbb{Z})^*$.

Use Lagrange's theorem to see that $|a| = |\langle a \rangle|$ must divide $\phi(n)$. Thus $a^{\phi(n)} \equiv 1 \pmod{n}$.

3.4 Isomorphism Theorems

1st Isomorphism Theorem: If $\phi : G \rightarrow H$ is a homomorphism, then $\ker \phi \trianglelefteq G$ and $G/\ker \phi \cong \phi(G)$.

2nd Isomorphism Theorem: Let $A, B \leq G$ and $A \leq N_G(B)$. Then $AB \leq G$, $B \trianglelefteq AB$, $A \cap B \trianglelefteq A$, and $AB/B \cong A/A \cap B$.



Lattice diagram of 2nd isomorphism theorem

3rd Isomorphism Theorem: Let $H, K \trianglelefteq G$ with $H \leq K$. Then $K/H \trianglelefteq G/H$ and

$$(G/H)/(K/H) \cong G/K$$

4th Isomorphism Theorem: Let $N \trianglelefteq G$. There's a bijection from the set of subgroups $A \leq G$ containing N onto the set of subgroups A/N of G/N .

I.e. every subgroup of G/N is of the form A/N for some $A \leq G$ containing N .

3.5 The Hölder Program

3.5.1 Composition Series

Call G a **simple group** if $|G| > 1$ and the only normal subgroups of G are 1 and G .

The sequence

$$1 = N_0 \leq N_1 \leq N_2 \leq \cdots \leq N_{k-1} \leq N_k = G$$

is called a **composition series** if $N_i \trianglelefteq N_{i+1}$ and N_{i+1}/N_i is a simple group. The quotient groups N_{i+1}/N_i are **composition factors** of G .

E.g. for the dihedral group D_8 ,

$$\begin{aligned} 1 &\trianglelefteq \langle s \rangle \trianglelefteq \langle s, r^2 \rangle \trianglelefteq D_8 \\ 1 &\trianglelefteq \langle r^2 \rangle \trianglelefteq \langle r \rangle \trianglelefteq D_8 \end{aligned}$$

3.5.2 Hölder Program

Jordan-Hölder Theorem: Let G be a finite group with $|G| > 1$. Then G has a composition series, and the composition factors are unique up to a permutation.

This suggests the **Hölder program**:

1. Classify all finite groups
2. Find all ways of combining two groups into one.

Theorem: There are 18 infinite simple group families and 26 simple groups not in those families (sporadic simple groups), such that every finite simple group is isomorphic to 1 in this list.

Feit-Thompson Theorem: If G is a simple group of odd order, then $G \cong Z_p$ for some prime p .

Proof: Both are trivial.

We call G **solvable** if it has the composition

$$1 = G_0 \trianglelefteq G_1 \trianglelefteq G_2 \trianglelefteq \cdots \trianglelefteq G_s = G$$

where G_{i+1}/G_i is Abelian.

3.6 The Alternating Group A_n

Call a 2-cycle $(i\ j)$ a **transposition**. Every element of S_n can be written as a product of transpositions.

Define the polynomial Δ and the permutation σ such that

$$\Delta = \prod_{1 \leq i < j \leq n} (x_i - x_j)$$

$$\sigma(\Delta) = \prod_{1 \leq i < j \leq n} (x_{\sigma(i)} - x_{\sigma(j)})$$

Since σ is a bijection, only a few terms $(x_{\sigma(i)} - x_{\sigma(j)})$ get their signs switched, so $\sigma(\Delta) = \pm\Delta$. Define ϵ such that

$$\epsilon(\sigma) = \begin{cases} +1 & \sigma(\Delta) = \Delta \\ -1 & \sigma(\Delta) = -\Delta \end{cases}$$

- E.g. for $\sigma = (1\ 2\ 3\ 4)$, we have $(1\ 2\ 3\ 4)(\Delta) = (-1)^3(\Delta)$, so $\epsilon((1\ 2\ 3\ 4)) = -1$. This is **not** the Levi-Civita symbol.
- $\epsilon(\sigma)$ is called the **sign** of σ .
- σ is an even permutation if $\epsilon(\sigma) = +1$, odd if $\epsilon(\sigma) = -1$.
- The map $\epsilon : S_n \rightarrow \{\pm 1\}$ is a homomorphism.

Theorem: transpositions $(i\ j)$ are all odd permutations.

Proof: Note that $\epsilon((1\ 2)) = -1$ since it only flips $(x_1 - x_2)$. For a general $(i\ j)$, let λ be the permutation exchanging 1 and i , 2 and j . Then $(i\ j) = \lambda(1\ 2)\lambda$. ϵ is a homomorphism, so

$$\epsilon((i\ j)) = \epsilon(\lambda(1\ 2)\lambda) = \epsilon(\lambda)^2 \epsilon((1\ 2)) = -1$$

Define A_n , the **alternating group of degree n** , to be the kernel of ϵ . I.e. the set of even permutations. $S_n - A_n$ is the set of odd permutations.

By the 1st isomorphism theorem, $S_n/\ker \epsilon \cong \{\pm 1\}$, and by Lagrange's theorem, $|A_n| = n!/2$. A_n will become useful in the study of polynomials.

4 Group Actions

4.1 Definitions

Let G be a group acting on set A . For all $g \in G$, the map $\sigma_g : A \rightarrow A$ defined by $\sigma_g : a \mapsto g \cdot a$ is a permutation of A .

For each action of G on A , there's an associated homomorphism $\phi : G \rightarrow S_A$ defined by $\phi(g) = \sigma_g$, called the **permutation representation** associated with a given action.

- Kernel of action: $\{g \in G \mid g \cdot a = a, \forall a \in A\}$
- Stabilizer of a in G : $G_a = \{g \in G \mid g \cdot a = a\}$
- An action is **faithful** if the kernel is the identity.

A **permutation representation** of G is any homomorphism of G into the symmetric group S_A for some nonempty set A .

- Given any nonempty set A and homomorphism $\phi : G \rightarrow S_A$, the action of G on A can be $g \cdot a = \phi(g)(a)$. The kernel of this action is the same as the kernel of ϕ .

Let G act on set A . The relation on A defined by $a \sim b$ iff. $a = g \cdot b$ for some $g \in G$ is an **equivalence relation**. The **equivalence class** is $\mathcal{C}_a = \{g \cdot a \mid g \in G\}$. Alternatively, the **orbit of G containing a** .

- The orbit of a : set of all elements a can move to under a group action.
- $|\mathcal{C}_a| = |G : G_a|$, index of stabilizer a
- Action of G on A is **transitive** if there's only 1 orbit. I.e. for all $a, b \in A$, there exists $g \in G$ such that $a = g \cdot b$.

Cayley's Theorem: Every group is isomorphic to a subgroup of some symmetric group. If G has order n , then G is isomorphic to a subgroup of S_n .

Proof: Let G be a group, and let A be the set of left cosets of $1 \in G$. Let G act by left multiplication on A . The homomorphism $\pi : G \rightarrow S_G$ is that of a group action. The kernel of π is

$$\begin{aligned} \ker \pi &= \{g \in G \mid gx1 = x1 \quad \forall x \in G\} \\ &= \{1\} \end{aligned}$$

Thus π is an isomorphism.

4.2 Groups Acting on Themselves by Conjugation

4.2.1 Definitions

Let G act on $A = G$ through conjugation, such that

$$g \cdot a = gag^{-1}, \quad \forall g, a \in G$$

$a, b \in G$ are conjugate in G if there exists a $g \in G$ such that $b = g \cdot a = gag^{-1}$. I.e. if a and b are in the same orbit of G acting on itself by conjugation. These orbits are called the **conjugacy classes of G** .

- For an Abelian group, the conjugacy class of each $g \in G$ is $\{g\}$.
- Subsets $S, T \subseteq G$ are conjugate in G if there exists a $g \in G$ such that $T = gSg^{-1}$. I.e. if S, T are in the same orbit of G acting on itself by conjugation.

For $S \subseteq G$, if G acts on S with $g \cdot a = gag^{-1}$, $\forall a \in S$, the number of elements in the orbit/equivalence class containing S is $|G : G_S|$.

$$G_S = \{g \in G \mid gSg^{-1} = S\} = N_G(S)$$

- The number of conjugates of a subset $S \subseteq G$ is the index of the normalizer of S , i.e. $|G : N_G(S)|$.
- The number of conjugates of an element $s \in G$ is the index of the centralizer of s , i.e. $|G : C_G(s)|$.

The Class Equation: Let G be a finite group, and let $g_1, g_2, \dots, g_r$ be the representatives of distinct conjugacy classes of G not contained in the center $Z(G)$. Then

$$|G| = |Z(G)| + \sum_{i=1}^r |G : C_G(g_i)|$$

Corollary: If P is a group of prime power p^α for some $\alpha \geq 1$, then P has a nontrivial center $Z(P) \neq 1$.

Proof: By the class equation,

$$|P| = |Z(P)| + \sum_{i=1}^r |P : C_P(g_i)|$$

and by assumption, $C_P(g_i)$ is non-central, so $|C_P(g_i)| \neq |P|$. p divides $|P : C_P(g_i)|$, so p must divide $|Z(P)|$ and $|Z(P)| \neq 1$.

4.2.2 Conjugacy in Symmetric Groups

Let $\sigma, \tau \in S_n$ and let σ have the cycle decomposition $(a_1 \ a_2 \ \dots \ a_{k_1})(b_1 \ b_2 \ \dots \ b_{k_2}) \dots$. Then $\tau\sigma\tau^{-1}$ has the cycle decomposition

$$(\tau(a_1) \ \tau(a_2) \ \dots \ \tau(a_{k_1}))(\tau(b_1) \ \tau(b_2) \ \dots \ \tau(b_{k_2})) \dots$$

Proof: If $\sigma(i) = j$, then $\tau\sigma\tau^{-1}(\tau(i)) = \tau(j)$. Thus if i, j appears in the cycle decomposition of σ , then $\tau(i), \tau(j)$ appears in the cycle decomposition of $\tau\sigma\tau^{-1}$.

If $\sigma \in S_n$ is the product of disjoint cycles of length $n_1, n_2, \dots, n_r$ with $n_1 \leq n_2 \leq \dots \leq n_r$, then $n_1, n_2, \dots, n_r$ are the **cycle type** of σ . Two elements of S_n are conjugate iff. they have the same cycle type.

The number of partitions of n equals the number of conjugacy classes of S_n . E.g. for $n = 5$,

Partition of 5	Representative of Conjugacy Class
1, 1, 1, 1, 1	1
1, 1, 1, 2	(1 2)
1, 1, 3	(1 2 3)
1, 4	(1 2 3 4)
5	(1 2 3 4 5)
1, 2, 2	(1 2)(3 4)
2, 3	(1 2)(3 4 5)

4.3 Automorphisms

An isomorphism of G onto itself is called an **automorphism**. The set of all automorphisms of G is $\text{Aut}(G)$.

Let $H \trianglelefteq G$. Then G acts by conjugation on H as an automorphism of H . I.e. conjugation by $g \in G$ is an automorphism of H . The permutation representation of the homomorphism is $\phi : G \rightarrow \text{Aut}(H)$ with kernel $C_G(H)$.

$$h \mapsto ghg^{-1}, \quad \forall h \in H$$

Let $g \in G$. Conjugation by g is an **inner automorphism** of G , and the subgroup of $\text{Aut}(G)$ containing all inner automorphisms of G is called $\text{Inn}(G)$.

- Note that $\text{Inn}(G) \leq \text{Aut}(G)$, so $\text{Inn}(G) \cong G/Z(G)$.

A subgroup $H \leq G$ is **characteristic in G** , denoted $H \text{ char } G$, if every automorphism of G maps H to itself: $\sigma(H) = H$ for all $\sigma \in \text{Aut}(G)$.

- Characteristic subgroups are normal.
- Although normality isn't transitive, if $K \text{ char } H$ and $H \trianglelefteq G$, then $K \trianglelefteq G$.
- Characteristic subgroups are like “strongly normal” subgroups.

Theorem: For any $H \leq G$, $N_G(H)/C_G(H)$ is isomorphic to a subgroup of $\text{Aut}(H)$. If $H = G$, then this becomes that $G/Z(G)$ is isomorphic to a subgroup of $\text{Aut}(G)$.

Proof: Note that $H \trianglelefteq N_G(H)$. Conjugation by $g \in G$ is an automorphism of H with permutation representation $\phi : N_G(H) \rightarrow \text{Aut}(H)$ with kernel $C_G(H)$. By the 1st isomorphism theorem,

$$N_G(H)/C_G(H) \cong \text{Subgroup of } \text{Aut}(H)$$

Theorem: The automorphism group of cycle groups of order n , Z_n , is isomorphic to $(\mathbb{Z}/n\mathbb{Z})^\times$, an Abelian group of order $\phi(n)$.

Proof: Let $Z_n = \langle x \rangle$. If $\psi \in \text{Aut}(Z_n)$, then $\psi(x) = x^a$ for some $a \in \mathbb{Z}$. Further, $|x| = n$ so a is defined mod n . We seek to show a bijection between Z_n and $(\mathbb{Z}/n\mathbb{Z})^\times$. Define the homomorphism (it can be shown that $\Psi(\psi_a \circ \psi_b) = \Psi(\psi_a) \circ \Psi(\psi_b)$)

$$\begin{aligned} \Psi : \text{Aut}(Z_n) &\rightarrow (\mathbb{Z}/n\mathbb{Z})^\times \\ \psi_a &\mapsto a \pmod{n}, \quad \psi_a \in \Psi \end{aligned}$$

This map is surjective. It is also injective, thus Ψ is an isomorphism.

4.4 Sylow Theorems

4.4.1 Statement

Let p be a prime. A group of order p^α for some $\alpha \geq 1$ is called a p -group. A subgroup of G that is a p -group is called a p -subgroup.

If $|G| = p^\alpha m$ where $p \nmid m$. A subgroup of order p^α (i.e. max subgroup of size that's a power of p) is called a **Sylow p -subgroup of G** .

- Set of Sylow p -subgroups of G : $\text{Syl}_p(G)$
- Number of Sylow p -subgroups of G : $n_p(G)$

Sylow Theorems: (partial converse to Lagrange's Theorem) Let $|G| = p^\alpha m$, where p is prime and $p \nmid m$. Then

1. $\text{Syl}_p(G) \neq \emptyset$
2. If P is a Sylow p -subgroup of G , and Q is any p -subgroup of G , then there exists $g \in G$ such that $Q \leq gPg^{-1}$. Further, any 2 Sylow p -subgroups of G are conjugate in G .
3. The number of Sylow p -subgroups of G is of the form $1 + kp$, i.e. $n_p \equiv 1 \pmod{p}$. Further, $n_p = |G : N_G(P)|$ for a Sylow p -subgroup P . Thus n_p divides m .

If P is a Sylow p -subgroup of G , the following are equivalent:

- 1) P is the unique Sylow p -subgroup of G , $n_p = 1$
- 2) $P \trianglelefteq G$
- 3) $P \text{ char } G$
- 4) All subgroups generated by elements of power p^n for some n are p -groups.

4.4.2 Applications

Theorem: If $|G| = pq$ for prime p, q with $p < q$, let $P \in \text{Syl}_p(G)$ and $Q \in \text{Syl}_q(G)$. Then $Q \trianglelefteq G$. If $P \trianglelefteq G$ as well, then G is cyclic.

Proof: (1) Since $Q \in \text{Syl}_q(G)$, $n_q = 1 + kq$ for some $k \geq 0$. And n_q divides p , so $p < q$ and $p = \text{prime}$ enforces $k = 0$. Thus $n_q = 1$, and this is equivalent to $Q \trianglelefteq G$.

Now since $P \in \text{Syl}_p(G)$, $n_p = 1 + kp$ for some $k \geq 0$. Because n_p divides q , either $n_p = 1$ or $n_p = q$. If $n_p = q$, then $q \equiv 1 \pmod{p}$. If $n_p = 1$, then q is not $1 \pmod{p}$, and $P \trianglelefteq G$.

- (2) If $P \trianglelefteq G$, then $n_p = 1$ so q is not $1 \pmod{p}$. Since P, Q have prime power, they're cyclic: $P = \langle x \rangle$ and $Q = \langle y \rangle$. If $P \trianglelefteq G$, $N_G(P) = G$, so $G/C_G(P) \cong \text{Subgroup of } \text{Aut}(P) = \text{Subgroup of } Z_p$. This has order $(p - 1)$, so by Lagrange's theorem,

$$\frac{|G|}{|C_G(P)|} = |Z_p| = p - 1$$

Neither p, q divide $p - 1$, so $G = C_G(P)$. Thus for $x \in P \leq Z_G(P)$, x commutes with all y , and $|xy| = pq$ or $G \cong Z_{pq}$.

E.g. Groups of Order 30: We show that all groups of order 30 have a normal subgroup isomorphic to Z_{15} .

Sketch: Any subgroup of order 15 is normal, since it has index 2. Also cyclic by previous result, since $15 = 3 \cdot 5$ and $3 \nmid (5 - 1)$. So we just show that there exists a subgroup of order 15.

Let $P \in \text{Syl}_p(G)$ and $Q \in \text{Syl}_q(G)$. Assume neither are normal. The possibilities then are $n_5 = 6$, $n_3 = 10$, not $n_5 = n_3 = 1$. There are $4 \cdot 6 = 24$ non-identity elements of order 5, and $2 \cdot 10$ non-identity elements of order 3. Total 44, yet $|G| = 30$, impossible. Thus at least 1 of P, Q must be normal, and

$$|PQ| = \frac{|P||Q|}{|P \cap Q|} = 15$$

5 Direct and Semidirect Products and Abelian Groups

5.1 Direct Products

The **direct product** $G_1 \times G_2 \times \cdots \times G_n$ of groups G_i , $1 \leq i \leq n$, with operations $\star_1, \star_2, \dots, \star_n$, is the set of n -tuples $(g_1, g_2, \dots, g_n)$ where $g_i \in G_i$ and the operation is defined componentwise:

$$(g_1, g_2, \dots, g_n) \star (h_1, h_2, \dots, h_n) = (g_1 \star_1 h_1, g_2 \star_2 h_2, \dots, g_n \star_n h_n)$$

And likewise for infinitely many n . For example, if $G_i = (\mathbb{R}, +)$, then $\mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}$ is the Euclidean n -space $\mathbb{R}^n$ such that

$$(a_1, a_2, \dots, a_n)(b_1, b_2, \dots, b_n) = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$$

- If $G_1, G_2, \dots, G_n$ are groups, their direct product is a group of order $|G_1||G_2| \cdots |G_n|$.
- The order of an element $x_1 x_2 \cdots x_n \in G$ is $\text{lcm}(|x_1|, |x_2|, \dots, |x_n|)$.
- For a fixed i , the set of elements of G with 1_{G_j} in the j -th position for all $i \neq j$ is a subgroup of G isomorphic to G_i .

$$G_i \cong \{(1, 1, \dots, 1, g_i, 1, \dots, 1) \mid g_i \in G_i\}$$

5.2 Fundamental Theorem of Finitely-Generated Abelian Groups

G is **finitely generated** if there's a finite subset A of G such that $G = \langle A \rangle$. For each $r \in \mathbb{Z}$ with $r \geq 0$, let $\mathbb{Z}^r = \mathbb{Z} \times \cdots \times \mathbb{Z}$ be the direct product of r $\mathbb{Z}$'s. Then $\mathbb{Z}^r$ is the **free Abelian group of rank r** .

Fundamental Theorem of Finitely Generated Abelian Groups: Let G be a finitely generated Abelian group. Then

$$G \cong \mathbb{Z}^r \times Z_{n_1} \times Z_{n_2} \times \cdots \times Z_{n_s}$$

for integers $r, n_1, n_2, \dots, n_s$ satisfying

1. $r \geq 0$ and $n_i \geq 2$ for all i .
2. n_{i+1} divides n_i for all applicable i .

The decomposition of G in this way is unique, called the **invariant factor decomposition of G** . r is the **free rank/Betti number** of G . Integers $n_1, n_2, \dots, n_s$ are the **invariant factors** of G .

The order of a finite Abelian group is the product of its invariant factors. G is said to be of type $(n_1, n_2, \dots, n_s)$, and s is the **rank of G** .

E.g. For an Abelian group of order 180, the possible invariant factors are

Invariant Factors	Abelian Groups
$2^2 \cdot 3^2 \cdot 5$	Z_{180}
$2 \cdot 3^2 \cdot 5, 2$	$Z_{90} \times Z_2$
$2^2 \cdot 3 \cdot 5, 3$	$Z_{60} \times Z_3$
$2 \cdot 3 \cdot 5, 2 \cdot 3$	$Z_{30} \times Z_6$

Theorem: Let G be an Abelian group of order $n > 1$ with unique prime factorization

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$$

Then $G \cong A_1 \times A_2 \times \cdots \times A_k$, with $|A_i| = p_i^{\alpha_i}$. For each $A \in \{A_1, \dots, A_k\}$ with $|A| = p^\alpha$,

$$A \cong Z_{p^{\beta_1}} \times Z_{p^{\beta_2}} \times \cdots \times Z_{p^{\beta_t}}$$

where $\beta_1 \geq \beta_2 \geq \cdots \geq \beta_t \geq 1$ and $\beta_1 + \beta_2 + \cdots + \beta_t = \alpha$. **This decomposition is unique.** p^{β_i} are the **elementary divisors of G** , and the decomposition is the **elementary divisor decomposition of G** .

I.e. G is the direct product of its Sylow p -subgroups.

E.g. For the Abelian group of order $n = 1800 = 2^3 \cdot 3^2 \cdot 5^2$,

Order p^β	Partitions of β	Abelian Groups
2^3	$(3), (2, 1), (1, 1, 1)$	$Z_8, Z_4 \times Z_2, Z_2 \times Z_2 \times Z_2$
3^2	$(2), (1, 1)$	$Z_9, Z_3 \times Z_3$
5^2	$(2), (1, 1)$	$Z_{25}, Z_5 \times Z_5$

Choose 1 Abelian group from each row. E.g. $G \cong Z_8 \times Z_3 \times Z_3 \times Z_{25}$. This **elementary divisor decomposition** can convert to **invariant factor decomposition** and vice versa.